Unit 1 – Social Media Analytics

1. Characteristics and Types of Social Media Analytics (SMA)

Definition

Social Media Analytics (SMA) is the process of collecting, measuring, analyzing, and interpreting data generated from social media platforms to understand users, trends, opinions, and business performance.

Characteristics of SMA

1. Large Volume of Data – Social media generates huge amounts of posts, comments, likes, shares, images, and videos.

2. Real-Time Analysis – Data can be monitored and analyzed almost immediately.

3. User-Generated Content – Most social media data is created by users through posts, reviews, comments, etc.

4. Unstructured Data – Social media contains text, images, videos, hashtags, links, and other forms of data.

5. Trend Identification – SMA helps identify popular topics, hashtags, products, and emerging trends.

6. Sentiment Analysis – It can determine whether people's opinions are positive, negative, or neutral.

7. Network-Based – It studies relationships and interactions between users.

Types of SMA

Social Network Analytics – Studies relationships and connections between users.

Social Media Text Analytics – Analyzes text such as comments, posts, and reviews.

Social Media Action Analytics – Studies actions such as likes, shares, clicks, and comments.

Social Media Hyperlink Analytics – Analyzes links shared between websites and social platforms.

Location Analytics – Studies geographical information associated with social media activities.

Search Engine Analytics – Analyzes search behavior, keywords, and search trends.

Example: A company can analyze Instagram comments to understand customer opinions about a newly launched product.

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2. SMA in Small Organization and Large Organization

Social Media Analytics can be used by both small and large organizations, but their

objectives and scale of implementation may differ.

SMA in Small Organizations

1. Customer Understanding – Helps small businesses understand customer preferences.

2. Marketing Analysis – Businesses can identify which posts and campaigns receive more engagement.

3. Competitor Analysis – They can monitor competitors' products and social media activities.

4. Low-Cost Promotion – Social media analytics helps measure the effectiveness of inexpensive digital marketing.

5. Customer Feedback – Reviews and comments can be analyzed to improve products and services.

Example: A small restaurant can analyze Instagram engagement to determine which food items are most popular.

SMA in Large Organizations

1. Large-Scale Data Analysis – Large companies analyze data from multiple social media platforms.

2. Brand Monitoring – They monitor their brand reputation across different regions.

3. Customer Segmentation – Customers can be divided according to interests, behavior, and demographics.

4. Campaign Measurement – Organizations can measure the performance of large advertising campaigns.

5. Predictive Analysis – Historical social media data can help identify possible future trends.

Example: A multinational company can analyze millions of social media interactions to understand customer opinions in different countries.

Difference

Small Organization Large Organization

Smaller amount of data Very large amount of data
Limited resources Dedicated analytics teams
Local customers Global customer base
Simple tools Advanced analytics platforms
Basic marketing analysis Complex customer and market analysis

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3. Seven Layers of Social Media

The seven layers of social media describe different aspects through which social media activities and data can be analyzed.

1. Social Network Layer

Focuses on the relationships and connections between users.

Example: Followers and friends on Facebook or Instagram.

2. Social Interaction Layer

Studies interactions between users.

Example: Likes, comments, replies, and shares.

3. Social Content Layer

Deals with the content created and shared by users.

Example: Posts, photos, videos, and blogs.

4. Social Communication Layer

Studies how information is communicated between users and communities.

Example: Conversations and discussions on social platforms.

5. Social Location Layer

Deals with the geographical information associated with social activities.

Example: A user posting from a particular location.

6. Social Search Layer

Focuses on searches, keywords, hashtags, and search trends.

Example: Finding how frequently users search for a particular product.

7. Social Analytics Layer

Uses collected social media data to generate useful insights for decision-making.

Example: Identifying customer sentiment toward a brand.

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4. SMA Cycle

The Social Media Analytics Cycle is a systematic process of collecting social media data and converting it into useful information.

Main Steps

1. Define Objectives
First, determine what needs to be analyzed.

Example: Find out why customers are dissatisfied with a product.

⬇️

2. Data Collection
Collect data from social media platforms such as posts, comments, likes, shares, and

reviews.

⬇️

3. Data Processing

Clean and organize the collected data by removing unnecessary or duplicate information.

⬇️

4. Data Analysis

Apply analytical techniques such as sentiment analysis, network analysis, and text analysis.

⬇️

5. Interpretation

Convert analytical results into meaningful insights.

⬇️

6. Decision Making

Organizations use the insights to make business or marketing decisions.

⬇️

7. Evaluation

Measure the results of the decisions and improve future strategies.

Simple Diagram

Objective → Data Collection → Data Processing → Analysis → Interpretation → Decision → Evaluation

Example: A company collects Twitter comments about its product, performs sentiment analysis, identifies negative feedback, and improves the product accordingly.

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5. Search Engine Analytics

Search Engine Analytics is the process of analyzing search-related data to understand what users search for and how they interact with search engines.

Important Components

1. Keyword Analysis – Identifies the keywords users frequently search.

2. Search Volume – Measures how frequently a keyword is searched.

3. Search Trends – Identifies changes in search interest over time.

4. Click Analysis – Studies which search results users click.

5. User Behavior – Helps understand what users are looking for.

6. Traffic Analysis – Determines how much website traffic comes from search engines.

7. Conversion Analysis – Measures whether search visitors perform desired actions.

Uses

Search Engine Optimization (SEO)

Understanding customer interests

Identifying trending topics

Improving website content

Measuring marketing campaigns

Example: An online clothing store can analyze search keywords to determine which types of clothes customers are currently searching for.

Unit 2 – Social Media Analytics

1. Nodes, Edges and Tie with Example

These are important concepts in social network analysis.

Node

A node represents an individual entity in a network.

Example: In a social media network, each user can be represented as a node.

Edge

An edge represents a connection between two nodes.

Example: If two users follow each other, the connection between them can be represented as an edge.

Tie

A tie represents the relationship or interaction between two connected nodes.

Example: Friendship between two Facebook users is a social tie.

Example

Suppose there are three users:

A – B – C

A, B and C = Nodes

Connections between A-B and B-C = Edges

Friendship/following relationship = Ties

Importance

Nodes, edges, and ties help analysts understand:

Who is connected to whom

Strength of relationships

Information flow

Important users

Structure of social networks

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2. Degree, Distribution, Density, Trust and Connectivity

1. Degree

The degree of a node is the number of connections it has with other nodes.

Example: If a user has 100 followers, the user's network degree may be represented by the number of connections.

2. Distribution

Distribution describes how connections or other network properties are spread among nodes.

Example: In a network, some users may have thousands of connections while others have only a few.

3. Density

Network density indicates how strongly connected the nodes in a network are.

A highly dense network has many connections between its nodes.

4. Trust

Trust represents the level of confidence users have in other users or information within a network.

Example: People may trust product recommendations from a person they regularly follow.

5. Connectivity

Connectivity refers to how nodes in a network are connected to one another.

High connectivity means information can travel through many possible paths.

Importance

These concepts help analyze:

Network structure

Influential users

Information flow

Relationships

Community structure

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3. Sources of Location Data and Categories of Location Analytics

Sources of Location Data

1. GPS

GPS-enabled smartphones and devices provide geographical coordinates.

2. Mobile Networks

Mobile towers can provide approximate information about a user's location.

3. Social Media Check-ins

Users can share locations through check-ins and location tags.

4. Wi-Fi

Wi-Fi networks can help determine approximate user locations.

5. IP Address

An IP address can provide approximate geographical information.

6. Geotagged Content

Photos, posts, and videos may contain geographical information.

Categories of Location Analytics

1. Descriptive Location Analytics

Describes where activities are occurring.

Example: Finding areas where most social media posts originate.

2. Spatial Analytics

Studies relationships and patterns between geographical locations.

3. Movement Analytics

Studies movement patterns of people or objects.

Example: Analyzing travel routes.

4. Predictive Location Analytics

Uses historical location data to predict future patterns.

Example: Predicting areas that may receive high customer traffic.

5. Real-Time Location Analytics

Analyzes location information as it is generated.

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4. Search Engine Tools

Search engine tools are software or online services used to analyze search behavior,

keywords, trends, and website performance.

Common Search Engine Tools

1. Google Trends – Used to analyze search trends and popularity of keywords over time.

2. Google Search Console – Provides information about website search performance.

3. Google Analytics – Helps analyze website traffic and user behavior.

4. Bing Webmaster Tools – Helps analyze website performance in Bing search.

5. Keyword Planner – Helps identify keywords and their search-related information.

Uses

Keyword research

SEO analysis

Website traffic analysis

Finding trending topics

Understanding user search behavior

Improving website visibility

Example: A business can use Google Trends to determine whether interest in a product is increasing or decreasing.

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5. Network Visualization

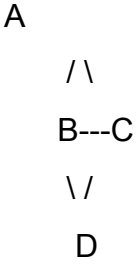
Network visualization is the graphical representation of relationships and connections between entities in a network.

In a network graph:

Nodes represent entities.

Edges represent relationships or connections.

Example



Here, A, B, C and D are nodes, while the lines represent connections.

Features

1. Shows relationships visually.

2. Helps identify highly connected nodes.

3. Helps identify communities or groups.

4. Shows information flow.

5. Makes complex networks easier to understand.

Uses

Social network analysis

Identifying influencers

Community detection

Studying communication patterns

Understanding relationships

Unit 3 – Social Media Analytics

1. Various Network Types in Social Media

Social media networks can be classified according to the type of relationships and interactions between users.

1. Friendship Network

Represents friendship relationships between users.

Example: Facebook friendship network.

2. Follower Network

Represents following relationships.

Example: Instagram or X follower network.

3. Communication Network

Represents direct communication between users.

Example: Messaging or email networks.

4. Collaboration Network

Represents users working together.

Example: Users collaborating on online projects.

5. Information Network

Represents the movement or sharing of information.

Example: Users sharing and reposting news.

6. Interest-Based Network

Connects users having similar interests.

Example: Online communities based on gaming, photography, or technology.

Importance

Network types help organizations understand:

User relationships

Communities

Information flow

Influencers

Common interests

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2. Network Analytics Tools

Network analytics tools are software applications used to analyze relationships and structures within networks.

1. Gephi

Gephi is used for network visualization and analysis. It helps identify important nodes and communities.

2. NodeXL

NodeXL is a network analysis tool that can be used to analyze social media networks.

3. UCINET

UCINET is used for social network analysis and provides various network analysis techniques.

4. Pajek

Pajek is designed for analyzing and visualizing large networks.

5. NetworkX

NetworkX is a Python library used to create, analyze, and visualize network structures.

Uses of Network Analytics Tools

Network visualization

Degree analysis

Community detection

Identifying influential users

Studying relationships

Understanding information flow

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3. Social Media Text Analytics

Social Media Text Analytics is the process of analyzing textual data generated on social media platforms.

Examples of text data include:

Posts

Comments

Reviews

Tweets

Messages

Hashtags

Main Techniques

1. Sentiment Analysis

Determines whether text expresses a positive, negative, or neutral opinion.

2. Keyword Analysis

Identifies frequently used and important words.

3. Topic Analysis

Identifies the main topics discussed by users.

4. Text Classification

Classifies text into predefined categories.

5. Entity Recognition

Identifies entities such as people, organizations, locations, and products.

Applications

Customer feedback analysis

Brand monitoring

Opinion analysis

Trend identification

Product improvement

Customer service

Example: A company can analyze thousands of product reviews to identify common complaints.

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4. Social Media Action Analytics

Social Media Action Analytics analyzes the actions and activities performed by users on social media platforms.

Common Social Media Actions

Likes

Comments

Shares

Clicks

Follows

Unfollows

Views

Reactions

Important Measures

1. Engagement

Measures how actively users interact with content.

2. Reach

Measures how many users are exposed to content.

3. Click-Through Rate

Measures how frequently users click a link or advertisement.

4. Conversion

Measures whether users complete a desired action.

5. Interaction Rate

Measures the level of interaction generated by content.

Applications

Measuring campaign performance

Understanding audience behavior

Improving content

Identifying popular posts

Measuring customer engagement

Example: A company can compare likes, comments, and shares on different Instagram posts to determine which content receives greater engagement.

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5. Social Media Hyperlink Analytics

Social Media Hyperlink Analytics is the analysis of hyperlinks shared between social media platforms, websites, and online content.

A hyperlink creates a connection between two web resources.

What It Analyzes

1. Number of Links – Measures how many links are shared.

2. Link Sources – Identifies where links originate.

3. Link Destinations – Determines where users are directed.

4. Link Sharing – Studies how frequently links are shared.

5. Referral Traffic – Measures website visitors coming through social media links.

6. Influence of Links – Identifies links or sources generating significant attention.

Applications

Website traffic analysis

Content promotion

SEO

Identifying influential websites

Understanding information flow

Tracking social media campaigns

Example

If a company shares a link to its website on Instagram and users click the link, hyperlink analytics can measure:

Instagram → Link → Company Website → User Visit

This helps the company understand how social media contributes to website traffic.